Symmetry: Even & Odd Functions **TEACHER KEY**

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Even function: symmetric across the <u>y-axis</u>; algebraically $f(-x) =$ <u>$f(x)$</u>.

Odd function: symmetric about the <u>origin</u>; algebraically $f(-x) =$ <u>$-f(x)$</u>.

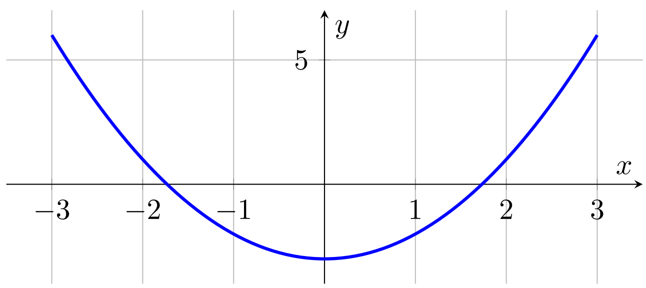
We do together. Is $f(x) = x^2 - 3$ even, odd, or neither?

Step 1. Replace x with $-x$: $f(-x) = (-x)^2 - 3 = \{\{x^2 - 3\text{—2cm}\}\}$.

Step 2. Compare to $f(x)$: they are <u>equal</u>, so $f(-x) = f(x)$.

Step 3. Therefore f is <u>even</u> (symmetric across the y-axis).

See it (even = y-axis mirror):



$y = x^2 - 3$ — the y-axis is a mirror line (even).

You Try. Solve each. Show every step, then name the answer.

a) Is $f(x) = x^3$ even, odd, or neither? odd: $f(-x) = -x^3 = -f(x)$

b) Is $f(x) = x^2 + x$ even, odd, or neither? neither

Summary (fill in): Compute $f(-x)$: equals $f(x) \Rightarrow$ <u>even</u>; equals $-f(x) \Rightarrow$ <u>odd</u>; otherwise <u>neither</u>.